

GradeSense — Graded Answer Paper

Question 1 — Electric Circuit Principles

Student: Ananya Rao · Roll: 24B · Subject: Science

2.5 / 5

Evaluated

100% confidence

GradeSense - Answer Sheet

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Q1. Science (5 marks)

An electric circuit is a close path in which electric current can flow. The battery gives the potentiall

diffrence which push the current around the circuit. Switch is used for open and close the circuit, if switch

is close then current will flow and bulb will glow, if switch is open then no current flow through the circuit.

In the circuit the battery, switch, resistor, bulb and ammeter is connected in series because current

pass through all of them one by one. Ammeter is connected in series because it meas ure the current

flowing in circuit. Voltmeter is also connected in the circuit to measure the potential diffrence, as shown

in diagram below -

When current pass through the circuit some of the current get used up by the bulb and resistor, that is why the bulb glow and become hot.

As per Ohms law, $V = IR$. So if we increase the resistance then the current flowing in the circuit will also

increase, because more resistance need more current to push through it. If the resistance is less then the

bulb will glow dim.

Answer 2 - English

Technology have changed the way we learn in a big way. Now students can find information easily on

internet, youtube videos, google etc. Before this student have to go library and read big books but now

everything is on phone.

I th ink technology make learning better because student can learn at there own pace and

1

2

explore different

topic which interest them. For example a student can watch video on youtube to understand hard topic.

But sometime student became too dependent on technology and they don't try to think on their own, they

just search google for answer.

In conclusion technology is good for learning but student should use it correctly.

Q.3 ECONOMICS

From the table we can make demand and supply graph, with quantity on x axis and price on y axis. Demand curve goes downward because when price increases people buy less, and supply curve goes upward because producer wants to sell more when price is high.

The two curves meet at price Rs 30 and quantity 60 units, this is called equilibrium point because at this

price the quantity demanded and quantity supplied is same.

If price is below equilibrium then demand is more than supply, this is called shortage, and price will start

increasing due to this. If price is above equilibrium also something will happen in the market.

If cost of production increases then producer will get more profit so they will supply more at the same

price, so the supply curve will shift to the right/upward, and new equilibrium will be at lower price and

higher quantity.

Rubric Breakdown

Score: 2.5 / 5

■ **Main circuit components (battery, switch, bulb, resistor) connected correctly in series** 1/1

The student correctly identifies that the main components (battery, switch, resistor, bulb) are connected in series.

■ **[1] Ammeter connected in series and voltmeter connected in parallel across the bulb (or component being** 0.5/1

The student correctly states that the ammeter is connected in series. However, the connection method for the voltmeter (parallel) is not explicitly stated in the text, and the diagram is missing.

■ **Explains closed path of current flow and functional roles of components** 1/1

The student correctly explains the concept of a closed path for current flow and the functional roles of the battery and switch. The explanation for the bulb's function is also present, though the phrasing 'current get used up' is a slight conceptual inaccuracy (energy is converted, not current consumed).

■ **[2] Explains Ohm's law relationship (increased resistance decreases current flowing when voltage is constant)** 0/1

The student correctly states Ohm's law formula ($V=IR$) but incorrectly explains the relationship between resistance and current. Increasing resistance *decreases* current (for a constant voltage), not increases it. Similarly, less resistance would lead to a brighter bulb, not dimmer.

■ **Properly labelled diagram with all components shown and conventional current direction indicated** 0/1

AI visual assessment (from the uploaded page image): The diagram shows all required components and they are labelled. However, the conventional current direction is not clearly indicated throughout the circuit; there is only a single ambiguous arrow pointing away from the circuit.